The librarian demo in MuJoCo: how contact force and deformation are simulated, what the skin panel plots, and when the skin helps

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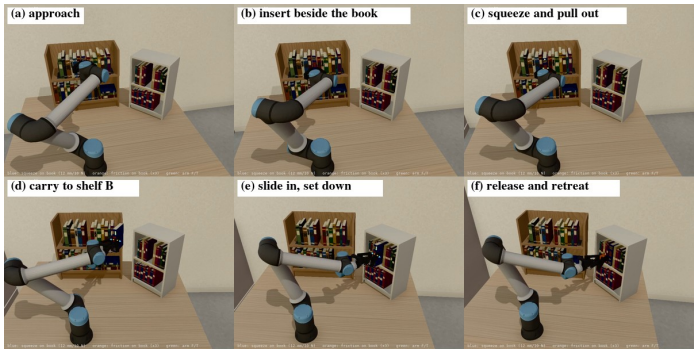


Figure 1: Key frames of the run (MuJoCo, Filament renderer). The blades slide into the 10 mm gaps beside the book (b); a 50 Hz controller squeezes, pulls the book out 21 cm and lifts it 8 cm (c); urx moves carry it to shelf B (d), slide it into the slot and set it down on a force threshold (e).

1 The demo, end to end

A UR10e with a Robotiq 2F-140 whose fingertips are our blades (`gripper_dohyun_v1`, a 0.5 mm PDMS layer on each flat face) takes a 40 mm book out of shelf A by approaching horizontally and shelves it in a 70 mm slot of shelf B, turned 45° toward the robot (Fig. 1). The task script uses only the `python-urx` calls of the real robot (`movel`, `speedl`, `get_tcp_force`, `gripper_action`, ...), so the same file drives the hardware. The video (`docs/media/librarian_demo.mp4`) has three synchronised panels: *left*, what the skin feels (Sec. 3); *centre*, MuJoCo rendered with the Filament physically based renderer plus a close-up that follows the tool, with arrows for the squeeze (blue) and friction (orange) on the book and the arm's wrench (green); *right*, the wrist camera and a urx console (joint angles, TCP pose and wrench, Robotiq registers, the last commands).

2 How MuJoCo produces force and displacement

2.1 Rigid bodies, joints and one step

The state is the generalised position q (joint angles, the pose of every free book, and the PDMS cells' displacements, Sec. 2.3) and velocity v. Each step of $h = 0.5$ ms solves

$$M(q)\,\dot{v} + c(q,v) = \tau + J(q)^{\top} f, \tag{1}$$

with M the mass matrix, c Coriolis/gravity, τ actuator and passive (spring, damper) forces, and f the *constraint forces*: contacts, the gripper's linkage (joint equalities) and joint limits, mapped by their Jacobian J [1]. Bodies are rigid: a book or a blade never strains. A **displacement** in MuJoCo is therefore a change of q — a body pose, a joint angle, or a degree of freedom we add on purpose to represent deformation (the PDMS layer).

2.2 Contact forces

Collision detection finds, for each pair of touching geoms, a contact frame (normal n, tangents t_1, t_2) and the signed distance r ($r < 0$: overlap). The contact force $f = (f_n, f_{t1}, f_{t2})$ must lie in the friction cone $\|f_t\| \le \mu f_n$, $f_n \ge 0$ (elliptic). MuJoCo computes all constraint forces of the step at once from a convex problem (Gauss's principle with *soft* constraints [2, 3]): the forces make each constraint's acceleration $a = J\dot{v}$ approach a reference

$$a^{\mathrm{ref}} = -b\,\dot{r} - k\,r, \tag{2}$$

a spring–damper set by `solref` (time constant, damping ratio), mixed with the unconstrained acceleration by an impedance $d \in (0,1)$ from `solimp` (regulariser $R = \frac{1-d}{d}\hat{A}$). Hence (i) contact forces are genuine forces in N: they balance gravity and actuation in Eq. (1) and obey action–reaction; (ii) contacts are slightly soft: at rest a small overlap r remains, about $\frac{1-d}{d}$ of the unresisted acceleration divided by k; (iii) friction is regularised: there is no static/kinetic difference and a sustained tangential load produces a slow creep (we keep it below 0.05 mm/s with a stiff friction-to-normal impedance ratio, `impratio` = 100; Sec. 6). Sensors are computed from the same solution: the UR10e's built-in wrist force/torque sensor is the wrench the flange transmits (MuJoCo `force`/`torque` sensors), minus the declared payload weight, as UR reports it.

2.3 The PDMS layer as a deformable body

MuJoCo has two ways to model deformable material [3]: *flex* elements (vertices are point masses; triangles or tetrahedra carry a Saint Venant–Kirchhoff finite-element elasticity with Young's modulus E and Poisson's ratio ν), and *lumped* models (rigid bodies joined by elastic joints or tendons). Our layer is $t = 0.5$ mm thick over 51×71 mm. We first built it as a 3D flex ($6 \times 8 \times 2$ vertices): on a fixed plate its shear stiffness was within 17% of the analytical value, but bolted to the moving finger its thin elements (aspect ratio 20:1 and more) made MuJoCo's effective-inertia solve ill-conditioned (contact and elastic forces no longer balanced). (The revision found the actual causes: two MuJoCo defects, Sec. 7.2.) We therefore use the *thin-layer limit* of linear elasticity, a leading-order approximation when t is much smaller than the loaded area [4] (its validity is quantified in Sec. 7.3): each column of PDMS deforms under its own traction only (a Winkler foundation). The layer is split into 5×7 cells of area $A = 10.2 \times 10.1\ \mathrm{mm}^2$; each cell is a point mass bonded to the blade by three slide joints (compression u_z, shear u_x, u_y) with

$$k_n = \frac{E(1-\nu)}{(1+\nu)(1-2\nu)}\,\frac{A}{t}, \qquad k_s = \frac{E}{2(1+\nu)}\,\frac{A}{t}, \tag{3}$$

(confined compression and shear of the column) and Kelvin–Voigt damping $\tau_d k$ (Fig. 2). With the placeholders $E = 1$ MPa (Sylgard-184-like [6]), $\nu = 0.45$, $\tau_d = 1$ ms: $k_n = 7.8 \times 10^5$ and $k_s = 7.1 \times 10^4$ N/m per cell. A 50 mg cell on such a spring oscillates at $\omega \approx 1.3 \times 10^5$ rad/s, so an explicit integrator would need $h < 2/\omega \approx 16\,\mu$s. MuJoCo's `discrete` integrator is implicit in position: it folds the stiffness into the effective inertia $\hat{M} = M + hD + h^2K$ and is stable at 0.5 ms; the layer then responds quasi-statically (in $\sim$1 ms), which is all a grasp needs. Each cell touches the object at one point (a flat ellipsoid filling the cell); that contact's own softness stands for the object's surface (a book cover), not for the PDMS.

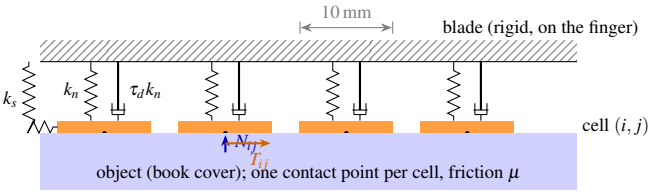


Figure 2: The PDMS layer as simulated: 5×7 columns of PDMS, each a point mass on three elastic slide joints (normal shown; two shear springs act along the face) bonded to the blade. Contact forces N_{ij}, T_{ij} come from MuJoCo's contact solver; the deformation u_{ij} is the cells' joint positions.

plotted quantity	how it is computed	status
N_{ij}, T_{ij} per cell	sum of the solver's contact forces on cell (i, j), in the PDMS frame	model physics
deformation u_{ij}	the cell's joint positions (Eq. 3)	placeholder E, ν
N, $\lvert T \rvert$, CoP, M	sums and moments over the cells	derived
margin q, Q	Eq. (4)	assumes $\mu = 1$
5×7 cells, 20 Hz	grid of the model; 50 ms frame averages	assumed sensor
noise, drift, hysteresis	none	not modelled
arm F/T	flange wrench − payload weight	as UR10e
slip, rotation	book pose relative to the TCP since grasp	sim only

Table 1: The skin panel, separated into what MuJoCo computes and what is assumed about the (uncharacterised) sensor. "Model physics": exact within the model, whose material values are placeholders.

3 What the skin panel plots

Table 1 separates what the physics model computes from what we *assume* about a sensor that is not characterised yet. The panel shows, per blade, a map of the contact force per cell (colour: normal N_{ij}; arrows: shear T_{ij}), a map of the layer's deformation (colour: compression; arrows: shear displacement), and time series of $N = \sum N_{ij}$, $|T| = |\sum T_{ij}|$, the margin q with the Robotiq force setting, and the arm's wrench with the book's slip. In the quasi-static limit the cell's elastic force equals its contact force ($k_s u_{xy} = T_{ij}$, $k_n u_z = N_{ij}$, checked to within 2% in our tests), so the two maps carry the same information here; a real sensor may report either, or strain, through its own unknown transfer function. The margin of each blade treats its contact patch as a whole, using an elliptical limit surface for sliding and twisting [7, 8]:

$$q = 1 - \sqrt{\left(\frac{|T|}{\mu N}\right)^2 + \left(\frac{|M|}{\mu N \rho}\right)^2}, \quad \rho = \tfrac{2}{3}\sqrt{A_c/\pi}, \qquad (4)$$

where $M = \sum (r_{ij} - c) \times T_{ij}$ is the twist of the shear field about the centre of pressure c and A_c the loaded area (ρ of a uniformly loaded disc). $q = 1$: no load; $q \le 0$: the patch slides or twists. The grasp margin is the weaker blade, $Q = \min(q_L, q_R)$. Without the twist term the book rotated out of the grasp: it is held near its spine, 65 mm from its centre of mass.

4 Control, and who decides what

Arm. Every urx call becomes joint targets for MuJoCo position servos with gravity compensation (as the UR controller does): `movej` by trapezoidal joint profiles; `movel` by a straight TCP line (and an angle–axis rotation) converted every 2 ms with damped least-squares inverse kinematics $\Delta q = J^\top (JJ^\top + \lambda^2 I)^{-1} e$ [9]; `speedl` by integrating a TCP twist through the same IK. The UR10e's published DH parameters are used, so poses match the real controller. **Gripper.** Robotiq registers: position 0–255, speed, and force 0–255 mapped to 10–125 N per finger (2F-140 data sheet) as the finger actuator's torque limit. **Decisions.** No learning or LLM agent is in this loop: the phases and their switches are hand-written rules (Table 2), a placeholder for MPC/MPPI or a learned policy with the same Observation→Command interface. Only the contact phase uses the skin. On the real robot the script stops for a typed confirmation before any motion.

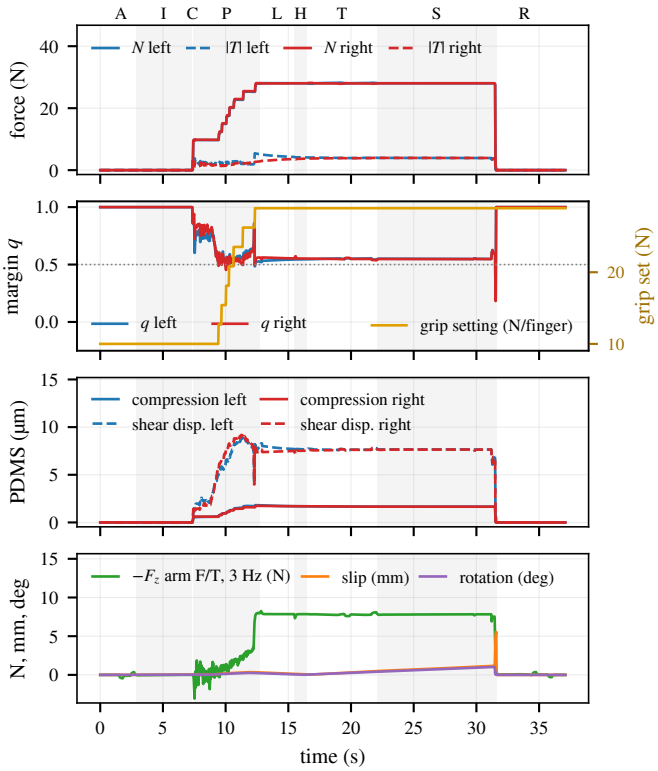


Figure 3: The demo run (0.8 kg book, $\mu = 1$, skin controller). Letters mark the phases: approach, insert, close, pull, lift, hold, carry (T), shelve (S), retreat. The controller raised the grip in steps until q settled at 0.55, just above $q^\star = 0.5$: 28 N per blade ($|T| = 3.9$ N). Holding, the PDMS compressed 1.7 μm and sheared up to 7.6 μm (a twist pattern: the book hangs 65 mm off the patch); the wrist F/T read the 7.8 N book. The book slipped 0.5 mm and 0.4° before set-down and landed 2 mm from the slot centre; neighbours moved 0.03 mm.

5 Ablation: when does the skin help?

Setup. The same task, 40 runs: five grip policies × book mass $\{0.4, 0.8, 1.6, 2.4\}$ kg × true PDMS–cover friction $\mu \in \{1.0, 0.5\}$. *skin*: the controller of Table 2, which always *assumes* $\hat{\mu} = 1$; *fixed*: the same motions at a constant force setting (10, 40, 70 or 125 N per finger) and no skin. A run succeeds if the book ends in shelf B's slot (centre within 15 mm, tilt and yaw within 5°). Successful runs placed the book within 7.7 mm, slipped ≤1.7 mm and rotated ≤1.4° in the grasp, and moved other books by ≤0.03 mm.

Results (Table 3, Fig. 4). *(i) When the load is not known in advance, the skin replaces a worst-case squeeze with a per-book one.* With the right μ, it shelved all four books with 15, 28, 54 and 81 N per blade; to match that without the skin a fixed grip must be tuned for the heaviest book (70–125 N on every book), and grips tuned lower (10, 40 N) dropped or rotated the heavier books. *(ii) It does not help — it hurts — when μ is overestimated.* With $\mu = 0.5$ the skin policy failed on every book, while a 125 N grip succeeded on all. This is structural: during sliding $|T| = \mu N$, so the computed margin $q = 1 - |T|/(\hat{\mu} N) = 1 - \mu/\hat{\mu}$ stays at $\ge q^\star = 0.5$ whenever $\mu \le \hat{\mu}(1 - q^\star)$, and the force is never raised; the abort never fires either. A force-ratio margin is only as good as its μ. The skin will help here only with slip detection from the shear field (incipient slip starts at the patch rim [4, 11, 10])

phase	signal	rule (who)
approach	—	`movel` 10 cm before the spines (script)
insert	wrist F/T	`movel` 12 cm, stop if $\lvert F\rvert > 15$ N (script)
close	skin N	close at 10 N/finger until both $N > 3$ N (controller)
pull, lift	skin Q	force setting +2.7 N per frame while $Q < 0.5$; arm pauses while $Q < 0.2$; abort if $Q < 0.2$ at full force for 1 s (controller)
carry	—	`movel` via a waypoint (script)
shelve	wrist F/T	stop on an unexpected 10 N; set down when the board pushes back 3 N; open (script)

Table 2: Phases, the signal that ends each one, and the rule. Thresholds are hand-tuned.

grip policy	shelved (of 4 masses) $\mu = 1.0$	$\mu = 0.5$	squeeze holding (N per blade)
skin, $\hat{\mu} = 1$	4	0	15 / 28 / 54 / 81
fixed 10 N	1	0	10
fixed 40 N	3	2	39
fixed 70 N	4	3	68
fixed 125 N	4	4	121

Table 3: Ablation summary (book mass 0.4 / 0.8 / 1.6 / 2.4 kg).

or an online estimate of μ: the next step for the controller, and a requirement (resolution, rate) for the sensor's characterisation.

6 Limitations

- **The sensor is not modelled.** The panel shows what the model computes, on its 5×7 grid, at an assumed 20 Hz, without noise, drift or hysteresis. Calibrating the real skin's output against these quantities is open work, as are its resolution and rate.
- **Material values are placeholders.** $E = 1$ MPa, $\nu = 0.45$ (real PDMS: ≈0.49, which makes the confined layer several times stiffer in compression and leaves shear unchanged), $\tau_d = 1$ ms (0 since the revision, Sec. 7.1), $\mu = 1$; all are arguments of `make_blade_assets.py`. Deformations are micrometres either way (Fig. 3).
- **Friction** is Coulomb with one μ, no static/kinetic difference, and residual creep (≈0.02–0.05 mm/s under the book's twist) after raising MuJoCo's `impratio` to 100. Load sharing among cells depends on the contact softness, so read the maps qualitatively; net forces are exact.
- **The task is scripted**: hand-tuned thresholds, known shelf and book poses (no perception), one book geometry, and every book rigid (its cover compliance is the contact softness).

7 Revision after the 29 October review

The review raised six points: the PDMS is 1.5 mm thick, so retry MuJoCo's flex and see whether it converges, and compare it with the cell model (item 1); linear elasticity only, without damping (2); how contact and the layer's boundary conditions are defined (3); a 3D robot-frame view of the skin (4); a revised report (5, this section); and when a high-level planner is needed (6, Sec. 8). A separate report [12] covers the flex in depth. All numbers below are from our runs unless a source is cited.

7.1 Model changes

The layer is now $t = 1.5$ mm thick: 0.5 mm stands proud of the blade face and 1.0 mm sits in a pocket. The PDMS frame is at mid-thickness. The material is linear elastic, with $E = 1$ MPa and $\nu = 0.45$ still placeholders, and has no Kelvin–Voigt or Rayleigh damping. The cell model of Sec. 2.3 therefore has $k_n = 2.62 \times 10^5$ and $k_s = 2.38 \times 10^4$ N/m per cell (Eq. 3 at $t = 1.5$ mm). Each cell's contact geom covers only the protruding 0.5 mm. Its contact time

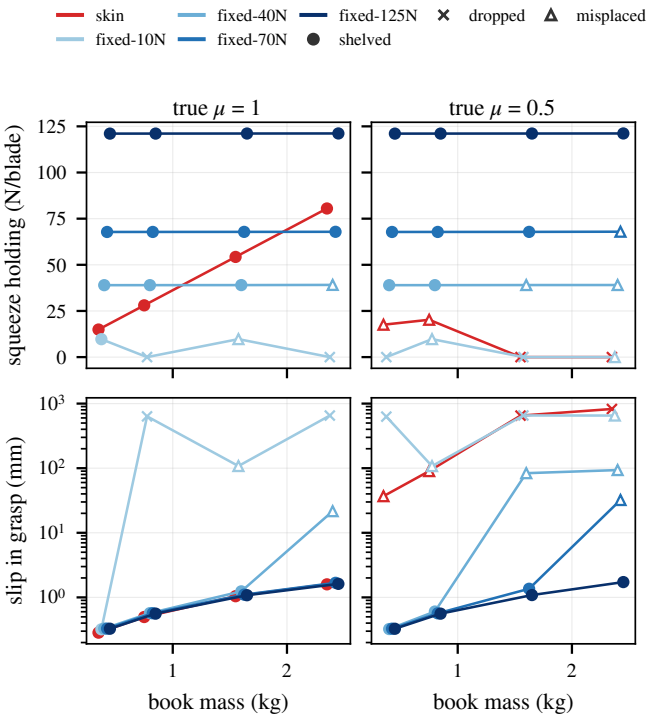


Figure 4: Ablation, 40 runs: squeeze while holding (top) and the book's slip in the grasp until set-down (bottom) against book mass, for the true friction $\mu = 1$ (left) and $\mu = 0.5$ (right). Filled markers: shelved; open: misplaced (△), dropped (×). The skin controller (red) assumes $\hat{\mu} = 1$.

constant is 3 ms, so a pressed object cannot sink to the blade. The constraint solver was also changed. With CG at its default cap of 100 iterations, 95% of the steps of the pull hit the cap, and the force balance on the book was off by up to 10.4 N ($m\mathbf{a}$ vs. the contact forces). Newton closes it to 3 mN at 9% more wall time, and the demo still shelves the book (1.8 mm centre error). The cell model now uses Newton. The flex needs CG (Sec. 7.2). The generator writes these options next to the PDMS model. Re-running the ablation of Sec. 5 with the revised cell model reproduces Table 3 in every outcome: the skin shelves 4 of 4 books at $\mu = 1$ and 0 of 4 at $\mu = 0.5$, and fixed grips of 10, 40, 70 and 125 N shelve 1, 3, 4 and 4 (and 0, 2, 3 and 4). The skin's squeeze per book is now 15, 28, 57 and 83 N per blade.

7.2 MuJoCo flex: verification and convergence

Two engine defects. MuJoCo's tetrahedral flex (`dof="full"`, Saint Venant–Kirchhoff) uses twice the Lamé λ. Patch tests with exact homogeneous strains give shear exactly, but uniaxial strain 1.82× and hydrostatic compression 1.93× too stiff at $\nu = 0.45$, for every mesh, ν and parent motion tested. The cause is one line of the compiler, which weights $(\mathrm{tr}\,E)^2$ by λ instead of $\lambda/2$. Entering $\nu_{\text{in}} = \nu/(2(1-\nu))$ and $E_{\text{in}} = 2G(1+\nu_{\text{in}})$ restores exact linear elasticity (patch ratios 1.000000), and the generator now does so. This, not volumetric locking as we first thought, made our earlier flex runs 1.65× too stiff in compression. Second, mounted on the finger, the flex diverged within three steps whenever any joint upstream was free. MuJoCo's per-island constraint solver leaves out the stiffness of a flex pinned to a jointed body (the monolithic solver includes it); `<flag island="disable"/>` fixes it. A stiff flex contact also needs a converged constraint solve (CG, 1000 iterations). Otherwise transient overloads turn the layer inside out, because its edge-length energy has no inversion barrier.

Table 4: Flex structures for the pad: error against the exact solutions (%) for the whole-face platen (no-slip) and the one-taxel punch, and wall time per 0.5 ms step. Robot: whole model with both pads; rig: pad alone. [a]Elastic operator only (no contact). [b]Mean ± 95% interval over 5 punch placements.

structure (vertices)	whole face	one taxel	ms/step
$11 \times 15 \times 3$, raw inputs (first try)[a]	+85	+345	–
$11 \times 15 \times 3$, remapped[a]	+4.4	+158	–
$6 \times 8 \times 2$, remapped, robot setting	+12	$+452 \pm 106$[b]	21
$41 \times 57 \times 3$, remapped, rig	−14	$+9 \pm 25$[b]	36–490
graded tet mesh under a known indenter	–	$+17 \pm 10$[b]	120–510

Neither defect has been reported upstream yet; drafts and minimal reproductions are in [12].

Convergence and cell size. With the remap, MuJoCo's elastic operator converges to the exact solutions (Sec. 7.3) from the stiff side, at about first order in the cell size: +2.6% for the whole face and +36% for a one-taxel punch at 1.3 mm cells. Contact adds a penalty spring in series with the layer, whose stiffness scales with the vertex mass. Refining through the thickness therefore makes the vertices lighter and the reading worse: whole face −12% with 2 elements, −36% with 6, each element costing 1.3–1.4×. More than two elements through the 1.5 mm, or in-plane cells below 2.5 mm for distributed loads, is overmeshing. Interpolated (`trilinear/quadratic`) flexes are exact in elasticity but unusable with contact: MuJoCo keeps 50 contacts per geom, and shear fails. Table 4 gives the chosen structures.

On the robot. The robot uses $6 \times 8 \times 2$ vertices, one element per taxel, with the remapped inputs, islands disabled, the stiff contact (time constant 0.1 ms) and CG with 1000 iterations. Whole-face loads, as in a grasp, are then within 12%. Taxel-scale loads are not resolved on this mesh; the fixed rig is for those. With these settings the gripper closes on a box and lifts it without divergence. Each pad carries the 2F-140's 120 N and the layer compresses 45 μm [12]. The full librarian task does *not* yet work with the flex. In one run (35 min wall time, 19× the cell model), the skins reported a friction margin of $q \approx 0.7$–0.8 during the pull, against 0.53 for the cell model under the same pull, so the controller squeezed only 23 N per blade instead of 28 N. When the book left the shelf, its twist reached the grasp. The book slipped 45 mm and rotated 31° within 0.2 s, and the task aborted during shelving. Two causes are identified. (i) A box collides with whole tetrahedra of the coarse $6 \times 8 \times 2$ mesh at their corners, and the contact forces gather near the pad centre. In a torsion test on a grasped box, their force-weighted radius is 14 mm against 19 mm for the cells (the exact pressure under a flat platen is nearly uniform, 23 mm). The box starts to turn at 0.58 N m against 1.11 N m for the cells at the same 24 N squeeze, and at 2.2 against 4.6 N m at 121 N. The exact pressure gives $\mu \sum N\bar{r} = 1.11$ N m, the cells' value, so the coarse flex resists half the twist the PDMS itself would: a contact artifact, not PDMS physics. A finer $11 \times 15 \times 2$ pad does not cure it (0.49 N m). At 23 N the flex's limit is about the book's 0.51 N m twist. The same concentration lowers the moment term of q; the force term ($|T|/\mu N \approx 0.1$) agreed with the cell model. (ii) MuJoCo's `mj_rnePostConstraint` skips contacts that involve a flex, so the simulated wrist force/torque sensor does not see any load carried by the skins (it read 0.03 N during the pull, against 3.2 N with cells). The insert and shelve guards are blind to the pads for the same reason. Fixes to try are a collision that samples the pad's surface vertices (MuJoCo does this only for planes), the moment term from the pad's own deformation, and a wrist wrench computed from the contact forces. Until then the demo of Secs. 2.3–5 keeps the cell model, with the errors of Table 5.

Table 5: Winkler stiffness / exact stiffness for the $51 \times 71 \times 1.5$ mm pad. Cases: rigid platen over the whole face (no-slip, frictionless) or over taxel rows 4–6; tangential platen; frictionless disc punch $a = 5$ mm; square punch on the centre taxel. Exact values from [12] (error bars < 0.1%).

case	$\nu = 0.45$	0.49	0.499
whole face, no-slip	1.07	1.22	2.08
whole face, frictionless	1.14	1.52	4.40
rows 4–6 (load edge)	1.09	1.35	3.12
tangential, whole face	1.005	1.005	1.005
disc punch, $a = 5$ mm	1.35	3.71	30.4
one taxel, no-slip	1.21	2.45	15.7
one taxel, frictionless	1.31	3.44	27.4

7.3 How good is the thin-layer (Winkler) model?

What [4] and [5] support. Johnson [4] treats the elastic foundation in §4.3 and layered bodies in §5.8. Secondary sources place his thin-layer treatment on pp. 138–141 [13]; we have not checked the book's own equation numbers. A layer bonded to a rigid base under a *uniform* traction has $w = pt/M$, i.e. our k_n [14]. Johnson argued that this remains a good approximation for varying tractions when t is small compared with the loaded area, and noted that as $\nu \to \frac{1}{2}$ the deformation involves in-plane flow of material [14]. The approximation keeps plane sections plane, which requires t much smaller than the contact half-width [13]. So [4] supports Eq. (3) as an *approximation*, not the claim in Sec. 2.3 that it is exact when t is small. Shull's review [5] does not support a local model: its flat-punch compliance, quoted by [15], grows as $(a/t)^3$ for $a \gg t$, the non-local confinement of a nearly incompressible layer. We keep [4] for Eq. (3) as an approximation and cite [5] only for confinement.

Validity criterion. For a layer bonded to a rigid base and loaded over a length ℓ, the thin-layer asymptotics of Chandler and Vella [16] give

$$w = \frac{p}{k_W} - \Gamma \frac{t^3}{G} \nabla^2 p, \qquad k_W = \frac{M}{t}, \tag{5}$$

where $\Gamma(\nu)$ depends on whether the surface is free of shear or bonded. The two terms are in the ratio $\chi = (1-2\nu)(\ell/t)^2$: Winkler holds for $\chi \gg 1$. In the incompressible limit ($\chi \ll 1$) the response is the non-local Poisson problem $-(t^3/3G)\nabla^2 p = w$ with $p = 0$ on the contact edge [17, 18]. For our pad, $\chi = 29$ (whole face, $\nu = 0.45$) but $\chi = 1.1$ for a load on one 10 mm taxel, and $\chi = 5.8$ and 0.22 at $\nu = 0.49$. Real Sylgard 184 has $\nu \approx 0.495$–0.4998 [19, 20, 6].

The gap, measured against exact solutions. Table 5 compares Winkler with ground truth computed without MuJoCo: an exact integral-transform solution of the bonded layer with a boundary-element contact solve for the local loads, and a converged quadratic finite-element model for the finite pad. The two agree to 0.12% where they overlap [12]. At the placeholder $\nu = 0.45$, Winkler is 7–14% too stiff for loads on the whole face and 21–35% for a taxel-sized punch. At $\nu = 0.49$ these become 22–52% and 2.5–3.7×, and at $\nu = 0.499$, 2–4× and 16–30×. Shear is local (K_s within 0.5%). The cell model is therefore usable only for whole-face loads at the placeholder Poisson ratio. It fails exactly where the skin carries information (taxel-scale loads, load edges) and for real PDMS. This is why the skin now moves to the flex.

7.4 How contact and boundary conditions are defined

In the literature. *Local* skin models decide contact and force independently at each point or cell from its own gap: penalty forces from the interpenetration depth at sampled tactile points [21, 22], MuJoCo's tactile sensor [23], tactile bodies on spring-damped joints like our cells [24], and hydroelastic contact, an elastic foundation of modulus E/H [25, 26]. Multi-body skins alter the contact geometry, and with compliant contacts the penetration depends on how many contact points the simulator detects, which can cause jumps [27]. *Superposition* models use an influence matrix calibrated by FEM [28, 29]. *Volumetric* FEM/MPM simulators fix the surface attached to the sensor, leave the contact face free, and define contact per surface node or element by penalty, barrier or complementarity [30, 31, 32, 33, 34, 35]; the nodes are coupled through the stiffness matrix. Against FEM on a flat elastomer, the displacement error of penalty models is about $3\times$ that of a propagation model [36, Table II]. MuJoCo's flex is volumetric; we found no published tactile pad built from it.

Boundary conditions of our layer. The displacement field needs a condition on every boundary: $\mathbf{u} = \mathbf{u}_{\text{blade}}$ on the bonded back face; traction-free lateral faces over the protruding 0.5 mm (the pocket walls around the inner 1.0 mm are not modelled); and on the front face, Signorini conditions ($g \geq 0$, $p \geq 0$, $gp = 0$) with Coulomb friction inside the contact and zero traction outside. In the flex these are exactly what is imposed: pinned back-face vertices, free lateral and front faces, and MuJoCo's contact rows on the free surface. The Winkler laws $p = k_W w$ and $\mathbf{T} = (G/t)\mathbf{u}$ have no in-plane derivatives, so they neither need nor admit edge conditions; Eq. (5) is second order in the plane and needs them ($p = 0$ at free and contact edges [17]).

The $+1,+2,+1$ question. In the cell model each cell had its own contact geom, decided contact from its own gap and moved as a rigid plate: cells displaced by $(1,2,1)$ carried $k_n(1,2,1)$, with steps in the surface at the cell borders and nothing enforcing continuity. In a continuum, and in the flex through shared vertices, the surface is continuous and contact is decided per surface point, so cells in contact need not deform equally, but their responses are coupled. The exact solution shows the coupling directly: pressing the centre taxel in by δ with a rigid no-slip square punch moves its four neighbours *outward*, by 0.035δ at $\nu = 0.45$ and 0.13δ at $\nu = 0.49$, because the nearly incompressible layer bulges around the punch [12]. A cell model represents neither the bulge nor the lateral flow that makes the pressed taxel 1.2–2.5$\times$ softer than its confined column (Table 5).

7.5 The skin panel in the robot frame

The 2D maps at the top of the video's skin panel (Sec. 3) are replaced by two 3D views in the UR10e base frame, both seen along the close-up camera's direction (Fig. 5). Each pad's 5×7 cells are drawn at their true pose, coloured by normal force, with the force of the book on each cell as an arrow (normal plus shear, 10 mm/N); the book is a wireframe, and the book's weight $m\mathbf{g}$ is drawn next to the sum of the pad forces. The second view shows each cell's displacement ($\times 1000$). Checks on a full run: the panel's per-cell forces sum to minus MuJoCo's contact force on the book to 6×10^{-14} N; while the book is held, the summed pad forces equal $-m\mathbf{g}$ within 0.19 N and 1.4°; the view matches MuJoCo's camera to 0.09 px RMS; a frame takes 0.14 s to draw. The old 2D panel is still available (`--panel 2d`).

8 When does the task need a high-level planner?

When. We call for planning when the next step is a choice between contact strategies rather than a gain adjustment. In Suh's framing, local smoothed models plus MPC absorb contact-mode switches, leaving a global planner for the non-local decisions [37, 38]. Tactile systems likewise handle slip inside a primitive and sequence primitives by graph search [39], stop a grip loop at a force cap [40], and answer rotational slip with a regrasp, not more force [41]. For us that moment is a guard of Table 2 firing (Table 6). The nominal scene needs no planner (Sec. 5). A heavier book needs one only once the abort fires at full force (below that the skin adapts, 15–81 N per blade); published remedies then change strategy, e.g. regrasp nearer the centre of mass [41] or pivot on the board [42], chosen by friction-cone and joint-torque feasibility as in [43]. A narrow gap trips the insert guard. The top-edge tilt is a documented sensed sequence (press $\approx$8 N, pull while rotating at constant force, close past 15°) [44], also run on a PR2 from procedurally generated plans [45]; the only VLM-planned bookshelf retrieval we found has no public code [46]. An overestimated μ triggers nothing (q stays high): slip detection or online μ must come first [47, 48].

Switching. `movel` tracking runs only while a secure-grasp predicate holds, checked every skin frame: grip force settled (as in Romano's StableContact [10]), $Q \geq q^\star$, no unexpected wrist force and (sim) no book–tool slip. The combination is ours, not a published rule. A violation, guard or abort calls the planner, as DoReMi replans and ReKep backtracks on a failed constraint check [49, 50] and a behaviour tree falls back on a failed condition [51]; meanwhile the grip is held and the arm frozen (Hi Robot's VLM instead replans on a 1 s timer or user input [52]). Recoveries run under force control or MPC, not open-loop tracking. This is our choice; Shirai et al. conclude that LQR on smoothed dynamics seems insufficient to stabilise contact-rich plans [53]. Control returns to tracking once the predicate holds.

Repositories. High level: Code as Policies [54] (Apache-2.0; simulated UR5e/Robotiq demo). An LLM writes Python that calls primitives in sensor-conditioned loops, like our urx script; its API would be `movel`, `speedl`, the gripper registers, the tactile controller and the guard outcomes, run only on planner calls. It adds no physics or monitor, and its notebooks call deprecated OpenAI models. Gemini Robotics-ER 2 could add image-grounded choices (which book, top edge, free slot), but is a preview model [55]. Primitives: judo, Python sampling MPC on MuJoCo [56]. Exact gradients through stiff or discontinuous contact can be biased or high-variance [57], and smoothed (sampled) gradients found descent directions where exact ones stalled [58]; the Pang–Suh global planner needs Drake [59]. Predictive sampling can only control what simulates much faster than real time, and its authors name soft bodies as a hard case [60]: flex stays the plant; rollouts use a rigid skin unless timing allows flex.

Shown vs. to test. The sources give the strategies and the switch pattern, not thresholds for our blades, a VLM with public code for the tilt, or a language-model planner that itself reads force or touch. To measure with flex: the abort mass, skin precursors of slip at low μ, the insert's gap limits, the tilt's force window, and grip safety during a $\sim$10 s planner call [61].

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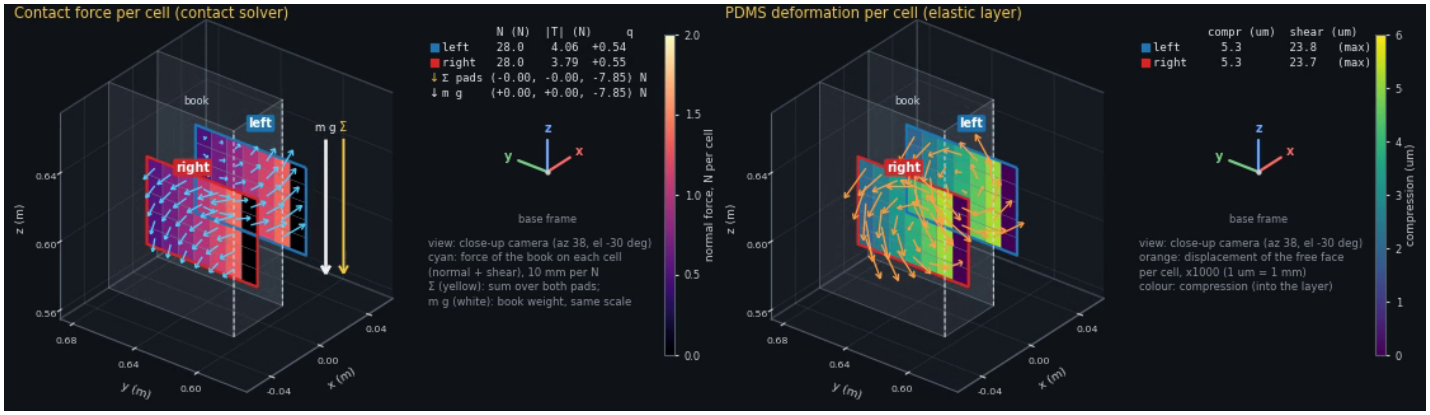


Figure 5: The new skin panel (top of the video's left column) while the book is held ($t = 16$ s): both pads' 5×7 cells at their pose in the UR10e base frame, seen along the close-up camera. Left: normal force per cell (colour) and the force of the book on each cell (arrows, 10 mm/N), with the book's weight $m\mathbf{g}$ (white) and the summed pad force Σ (yellow). Right: compression (colour) and free-face displacement (arrows, $\times 1000$).

case	our signal	level	what the literature does
gaps secured (now)	guards silent; $Q \geq q^\star$	none	scripted phases suffice (Sec. 5)
heavier, within 125 N	force setting rises, Q recovers; wrist F_z = weight	LL	grip raised on slip [10]; ours 15–81 N
heavier than grip or twist allows	abort: $Q < 0.2$ at full force for 1 s; sim: rotation	HL	regrasp nearer the CoM [41]; pull, regrasp, pull [62]; pivot on the support [42]; slide to a low-torque pose [63]; pick by force feasibility [43]
μ over-estimated	none on hardware (q blind); sim: book–tool slip	LL first	slip detection, online μ [47, 48, 40]; an LLM μ prior, set once, would share the blind spot [61]
gap too narrow	insert guard: wrist $\lvert F\rvert > 15$ N before 12 cm	HL	sideways force search [62]; compliant retries [64]; top-edge tilt [44, 45]; edge tilt by hybrid force–velocity control [65]; bookshelf planners [66, 67]; push-to-expose learned in MuJoCo [68]; VLM-chosen skill, no code found [46]
jam in pull-out; slot B blocked	pause and abort during pull, arm F/T; shelve guard (10 N)	HL	abort and replan [49]; explain and correct [69]; push items aside under F/T [70]
which book, top edge, free slot	none yet (poses known); wrist camera	VLM	keypoints on marked images [71]; pointing, success checks [55]; ask when unsure [72]

Table 6: When the high level (HL) is needed. LL: the existing low level suffices; VLM: a vision-language high level. Signals are those of Table 2; the case boundaries are untested beyond the ablation.

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